Lecture plan · for the lecturer

Recursion and induction

Programming · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Explain the call stack.
- Argue termination.
- Prove by induction.

Activity: Be the stack

1. Warm-up (5 min): Show the failing factorial and ask pairs to find the bug in one minute.
2. Act (15 min): Five students act as frames for fact(4), passing results back up.
3. Trace (15 min): Pairs trace fib(4) and count calls.
4. Prove (20 min): Write the proof of fact(n) = n! together, then one for list length.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- What does each frame remember?
- Why is fib slow?
- What may you assume?

Exit ticket

- Write the inductive hypothesis for a recursive list sum.

Try it in your next lecture
Line up five students as stack frames for fact(4); each writes their n on a card and passes the result back, then the class writes the induction proof together.

Seminar sheet · for students

Recursion and induction

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Name _____ Group _____ Date _____

1 **In the inductive step, what do you assume?**

◯ The smaller call is correct ◯ The function is correct for every n ◯ The current call is correct

2 **Count the calls.**

Naive recursive fib(5), with fib(0) = 0 and fib(1) = 1, returns _____ and makes _____ calls in total.

Word bank: 9 · 15 · 5 · 8

3 **Which does a correct recursive function need? Choose all that apply.**

☐ A base case that is reachable ☐ A correct combination of the smaller result

☐ A global counter ☐ A measure that decreases each call

4 **In the inductive step you may assume the function is correct for the input you are proving.**

◯ True ◯ False

5 **Put the proof of fact(n) = n! in order.**

☐ Expand fact(k+1) = (k+1) × fact(k)

☐ Prove fact(0) = 1 = 0!

☐ Conclude (k+1) × k! = (k+1)!

☐ Assume fact(k) = k!

Answers · for the lecturer

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1 **The smaller call is correct**

Assume only the smaller call, then prove the current one.

2 **Naive recursive fib(5), with fib(0) = 0 and fib(1) = 1, returns 5 and makes 15 calls in total.**

Calls satisfy C(n) = C(n−1) + C(n−2) + 1 with C(0) = C(1) = 1, giving 15; the exponential growth motivates memoisation.

3 **A base case that is reachable; A measure that decreases each call; A correct combination of the smaller result**

Global state is not needed and often causes bugs.

4 **False**

That would be circular; you assume only smaller inputs.

5 **Top to bottom: 3, 1, 4, 2**

Base case, hypothesis, step, conclusion.

For you to think about

Which mental model of recursion do your weakest students hold, and how would you find out?